

Lecture-4

Elementary Row Operations

Consider an $m \times n$ matrix

$$\begin{array}{l} R_1 \cdots \cdots \cdots \\ R_2 \cdots \cdots \cdots \\ \vdots \\ R_m \cdots \cdots \cdots \end{array} \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

By αR_i , we shall mean multiplying the i^{th} row with a number α .

By $R_i + \alpha R_j$, we shall mean adding i^{th} row R_i to αR_j .

Following three operations are called Elementary Row operations.

① Replacing R_i with $R_i + \alpha R_j$ for some number α .

Notation: $R_i \longrightarrow R_i + \alpha R_j$

② Replacing R_i with αR_i , Notation: $R_i \longrightarrow \alpha R_i$

③ Switching the rows R_i and R_j . Notation $R_i \longleftrightarrow R_j$

Examples

$$1. \quad \begin{pmatrix} 0 & 1 & 2 \\ 3 & 1 & 4 \\ 4 & 2 & 1 \end{pmatrix} \xrightarrow{R_1 \longleftrightarrow R_3} \begin{pmatrix} 4 & 2 & 1 \\ 3 & 1 & 4 \\ 0 & 1 & 2 \end{pmatrix}$$

$$2. \begin{pmatrix} 4 & 2 & 1 \\ 3 & 1 & 4 \\ 0 & 1 & 2 \end{pmatrix} \xrightarrow{R_2 \rightarrow R_2 + \frac{-3}{4} R_1} \begin{pmatrix} 4 & 2 & 1 \\ 0 & -\frac{1}{2} & \frac{13}{4} \\ 0 & 1 & 2 \end{pmatrix}$$

$$3. \begin{pmatrix} 4 & 2 & 1 \\ 0 & -\frac{1}{2} & \frac{13}{4} \\ 0 & 1 & 2 \end{pmatrix} \xrightarrow{R_1 \rightarrow \frac{1}{4} R_1} \begin{pmatrix} 1 & \frac{1}{2} & \frac{1}{4} \\ 0 & -\frac{1}{2} & \frac{13}{4} \\ 0 & 1 & 2 \end{pmatrix}$$

These row operations that we have performed are actually matrix multiplications.

$$\textcircled{1} \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 & 2 \\ 3 & 1 & 4 \\ 4 & 2 & 1 \end{pmatrix} = \begin{pmatrix} 4 & 2 & 1 \\ 3 & 1 & 4 \\ 0 & 1 & 2 \end{pmatrix}$$

$$\textcircled{2} \begin{pmatrix} 1 & 0 & 0 \\ -\frac{3}{4} & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 4 & 2 & 1 \\ 3 & 1 & 4 \\ 0 & 1 & 2 \end{pmatrix} = \begin{pmatrix} 4 & 2 & 1 \\ 0 & -\frac{1}{2} & \frac{13}{4} \\ 0 & 1 & 2 \end{pmatrix}$$

$$\textcircled{3} \begin{pmatrix} \frac{1}{4} & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 4 & 2 & 1 \\ 0 & -\frac{1}{2} & \frac{13}{4} \\ 0 & 1 & 2 \end{pmatrix} = \begin{pmatrix} 1 & \frac{1}{2} & \frac{1}{4} \\ 0 & -\frac{1}{2} & \frac{13}{4} \\ 0 & 1 & 2 \end{pmatrix}$$

Solving System of equations using elementary row operations

Examples

Solve the system of linear equations

①

$$2x + 3y = 8 \quad \text{--- a}$$

$$5x + 4y = 13 \quad \text{--- b}$$

First write this system in the following "augmented matrix form"

$$\begin{pmatrix} \overset{x}{2} & \overset{y}{3} & | & 8 \\ 5 & 4 & | & 13 \end{pmatrix}$$

Now we perform certain row operations

① $R_1 \rightarrow \frac{1}{2} R_1$

$$\begin{pmatrix} 2 & 3 & | & 8 \\ 5 & 4 & | & 13 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & \frac{3}{2} & | & 4 \\ 5 & 4 & | & 13 \end{pmatrix}$$

$$2x + 3y = 8$$

$$5x + 4y = 13$$

↓

$$x + \frac{3}{2}y = 4$$

$$5x + 4y = 13$$

② $R_2 \rightarrow R_2 + -5R_1$

$$\begin{pmatrix} 1 & \frac{3}{2} & | & 4 \\ 5 & 4 & | & 13 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & \frac{3}{2} & | & 4 \\ 0 & -\frac{7}{2} & | & -7 \end{pmatrix}$$

$$x + \frac{3}{2}y = 4$$

$$5x + 4y = 13$$

↓ mul Eqn ①
by -5

$$-5x - \frac{15}{2}y = -20$$

$$5x + 4y = 13$$

$$x + \frac{3}{2}y = 4$$

$$-\frac{7}{2}y = -7$$

⇐ add

$$\textcircled{\text{iii}} \quad R_2 \longrightarrow -\frac{2}{7} R_2$$

$$\left(\begin{array}{cc|c} 1 & \frac{3}{2} & 4 \\ 0 & -\frac{7}{2} & -7 \end{array} \right) \longrightarrow \left(\begin{array}{cc|c} 1 & \frac{3}{2} & 4 \\ 0 & 1 & 2 \end{array} \right)$$

$$\begin{aligned} x + \frac{3}{2}y &= 4 \\ -\frac{7}{2}y &= -7 \end{aligned}$$

$\Downarrow$

$$\begin{aligned} x + \frac{3}{2}y &= 4 \\ y &= 2 \end{aligned}$$

$$\textcircled{\text{iv}} \quad R_1 \longrightarrow R_1 + \frac{3}{2} R_2$$

$$\left(\begin{array}{cc|c} 1 & \frac{3}{2} & 4 \\ 0 & 1 & 2 \end{array} \right) \longrightarrow \left(\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & 2 \end{array} \right)$$

$$\begin{aligned} x + \frac{3}{2}y &= 4 \\ y &= 2 \end{aligned}$$

$\Downarrow \xrightarrow{\frac{3}{2} R_2}$

$$\begin{aligned} x + \frac{3}{2}y &= 4 \\ -\frac{3}{2}y &= -3 \end{aligned}$$

$$x = 1 ; y = 2 \quad \xleftarrow{R_1 - \frac{3}{2} R_2}$$

Solution : $x = 1 ; y = 2$